INCH-POUND
MS27480H
w/AMENDMENT 1
10 September 2024
SUPERSEDING
MS27480H
27 June 2012

DETAIL SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, STRAIGHT, CRIMP TYPE, BAYONET COUPLING, SERIES II

Inactive for new design after 27 June 2012.
For new design, use MS27473.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-38999.

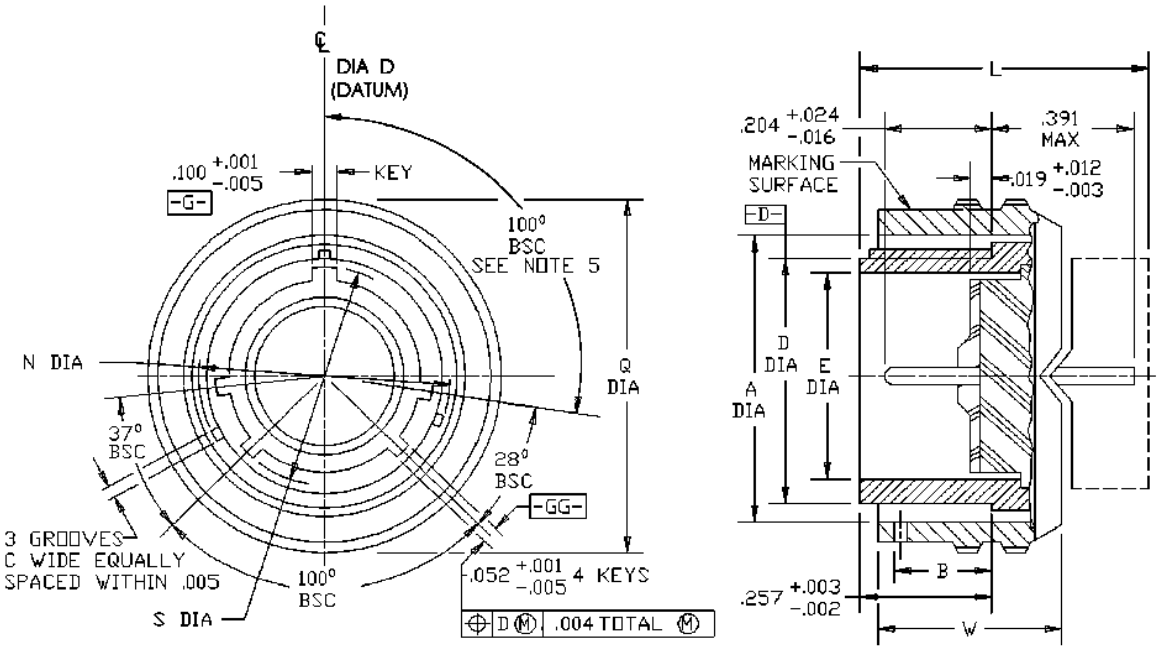


FIGURE 1. Plug, classes E and T.

AMSC N/A FSC 5935
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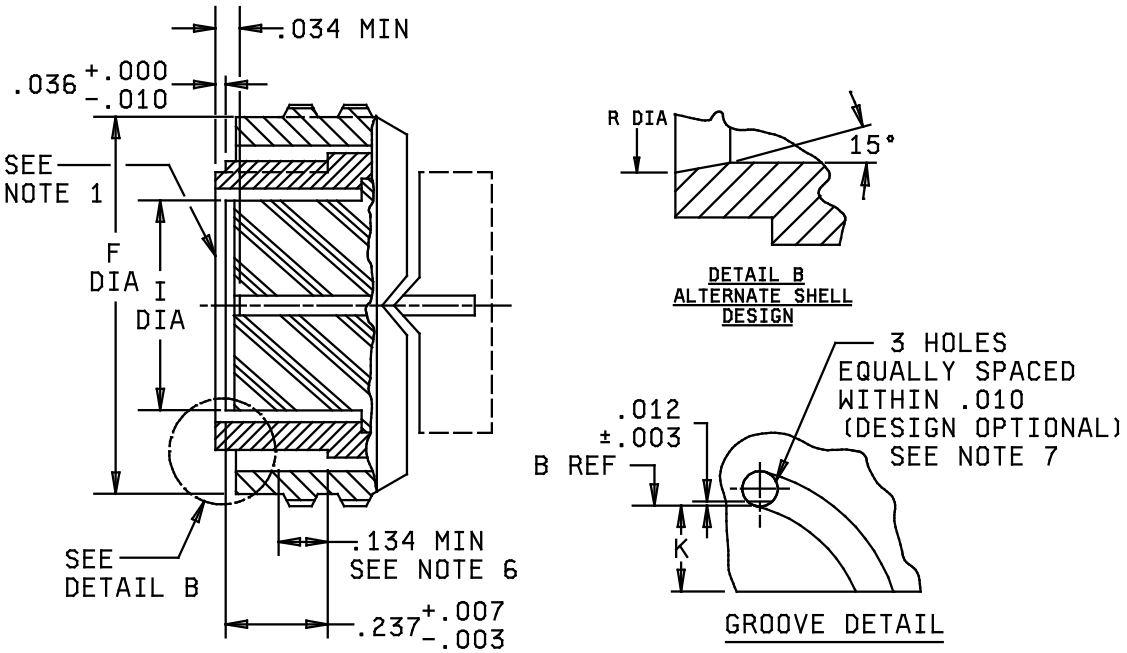


FIGURE 1. Plug, classes E and T – Continued.

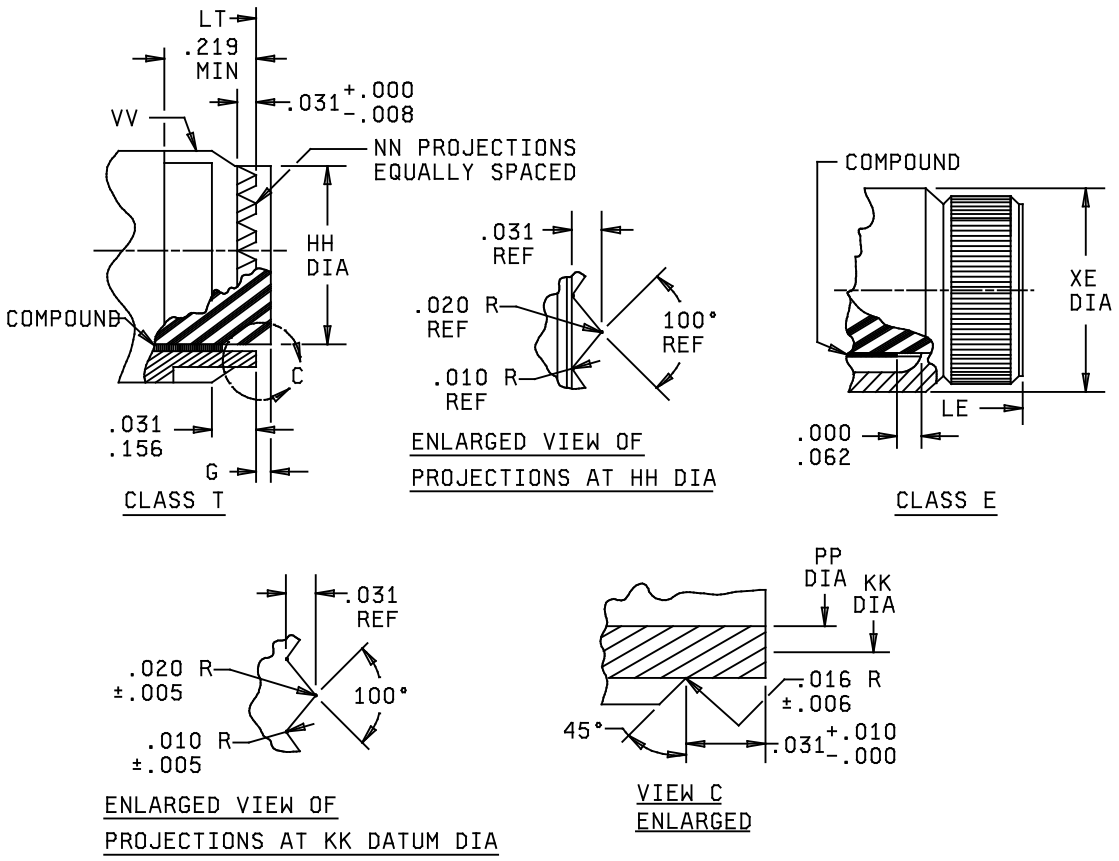
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Shell size	A dia +.005 -.001	B ±.020	C +.015 -.001	D dia +.001 -.005	E dia min	F dia +.000 -.006	K max REF	N dia +.010 -.006
8	.481	.096	.090	.357	.306	.630	.155	.576
10	.602			.485	.423	.752		.697
12	.761			.597	.537	.925		.871
14	.885			.722	.662	1.050		.995
16	1.010			.847	.787	1.172		1.120
18	1.136			.947	.876	1.304		1.245
20	1.260	.123	.137	1.072	1.001	1.435	.109	1.370
22	1.385			1.197	1.126	1.560		1.495
24	1.510			1.322	1.251	1.688		1.624

Shell size	Q dia max	R +.005 -.006 dia	S +.001 -.010 dia	W ±.015	For insert arrangement, see MS drawing	I dia max
8	.750	.316	.402	.547	27343, 27516	.290
10	.859	.436	.530		27344, 27517	.418
12	1.031	.549	.679		27345, 27518	.532
14	1.156	.675	.804		27346, 27519	.657
16	1.281	.799	.929		27347, 27520	.782
18	1.406	.893	1.029		27348, 27521	.871
20	1.531	1.018	1.154	.524	27349, 27522	.996
22	1.641	1.143	1.279		27350, 27523	1.121
24	1.766	1.268	1.404	.575	27351, 27524	1.246

FIGURE 1. Plug, classes E and T – Continued.

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NOTES:

1. Insert front surface shall be flat within .005 T.I.R.
2. Diameter D with respect to diameter S shall be concentric at MMC.
3. Diameters A and N shall be concentric within .008 T.I.R.
4. Dimensions are in inches.
5. Normal key position. For other key positions, see MIL-DTL-38999.
6. The point at which a gage pin, having the same basic diameter as the mating contact and a square face, touches socket contact spring.
7. Holes shall be provided for visual inspection of lock when mated with receptacle.

FIGURE 1. Plug, classes E and T - Continued.

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Shell size	G insert projection (class T only)	HH dia +.000 -.005	KK dia (DATUM)	NN	PP dia ± .003
8	.120 ± .030	.387	.366	12	.344
10		.515	.487	16	.472
12		.628	.609	20	.586
14		.754	.731	24	.711
16		.879	.853	28	.836
18		.985	.974	32	.942
20		1.110	1.097	36	1.067
22		1.235	1.218	40	1.192
24	.0900 ± .050	1.360	1.340	44	1.317

Shell size	LE max overall length	LT max overall length	XE dia max	VV thread UNEF-2A MOD (plated)	
				Size	Major dia MOD
8	.953	.766	.594	7/16 – 28	.421 – .417
10			.719	9/16 – 24	.542 – .538
12			.844	11/16 – 24	.667 – .663
14			.969	13/16 – 20	.791 – .787
16			1.094	15/16 – 20	.916 – .912
18			1.219	1 1/16 – 18	1.034 – 1.030
20			1.344	1 3/16 – 18	1.158 – 1.154
22			1.469	1 5/16 – 18	1.283 – 1.279
24		.844	1.594	1 7/16 – 18	1.408 – 1.404

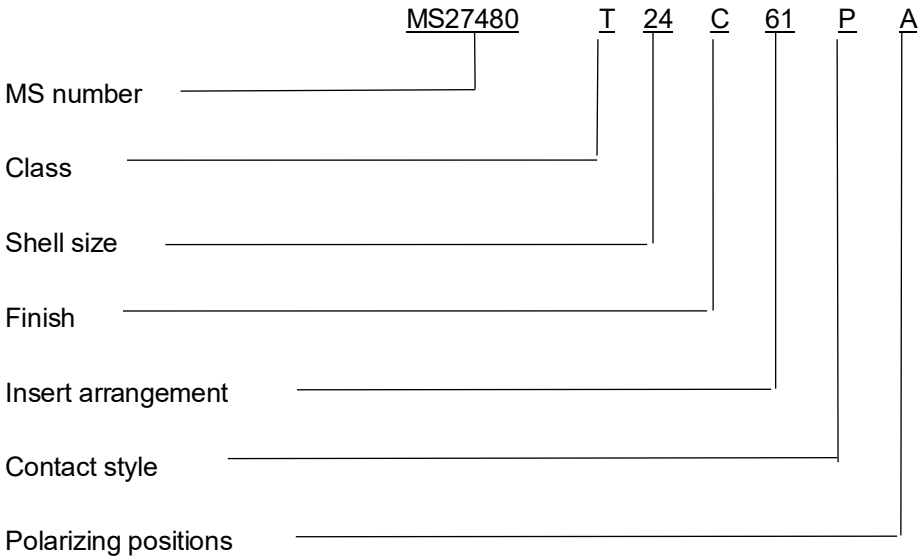
FIGURE 1. Plug, classes E and T - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.
Interface dimensions shall conform to MIL-DTL-38999.
Mating connectors: MIL-DTL-38999/10, MS27472, MS27474, MS27475, MS27476, MS27477, MS27478, MS27479, MS27480, MS27481, MS27497, MS27499, MS27504, MS27508 and MS27513.
Insert arrangements: See MIL-STD-1560.

Part or Identifying Number (PIN) example:



No letter is required for normal.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-38999, this document references the following:

MIL-DTL-38999/10
MIL-STD-1560
MS27472
MS27473
MS27474
MS27475
MS27476
MS27477
MS27478
MS27479
MS27480
MS27481
MS27497

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MS27499
MS27504
MS27508
MS27513

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

(Project 5935-2024-166)

Review activities:
Army – AR, MI
Navy – EC, MC, OS
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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